

Spring 2021 Real Analysis II — Homework 2

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1. Let $X := \{x \in C((0, 1]) : \|x\| < \infty\}$ where $\|x\| := \sup_{0 < t \leq 1} |x(t)|$. Show that $\|\cdot\|$ is a norm on X .

Proof. It is easy to verify that $\|\cdot\|$ is positive definite and homogeneous. Moreover,

$$\|x + y\| = \sup_{0 < t \leq 1} |x(t) + y(t)| \leq \sup_{0 < t \leq 1} |x(t)| + \sup_{0 < t \leq 1} |y(t)| = \|x\| + \|y\|.$$

Hence $\|\cdot\|$ is a norm. □

2. Let $X = C^1([-1, 1])$ and $\|x\| := (\int_{-1}^1 (|x(t)|^2 + |x'(t)|^2) dt)^{1/2}$ for any $x \in X$. Show that $\|\cdot\|$ is a norm on X , but X is not complete under $\|\cdot\|$. [Hint: consider $x_k(t) = (t^2 + \frac{1}{k^2})^{1/2}$ for $-1 \leq t \leq 1$ and $k \in \mathbb{N}$ and $x(t) = |t|$.]

Proof. It is easy to verify that $\|\cdot\|$ is positive definite and homogeneous. Moreover,

$$\begin{aligned} \|x + y\|^2 &= \int_{-1}^1 |x(t) + y(t)|^2 + |x'(t) + y'(t)|^2 dt \\ &= \int_{-1}^1 (|x(t)|^2 + |y(t)|^2 + |x'(t)|^2 + |y'(t)|^2 + 2(x(t)y(t) + x'(t)y'(t))) dt \\ &\leq \|x\|^2 + \|y\|^2 + 2 \int_{-1}^1 |(x(t), x'(t))| \cdot |(y(t), y'(t))| dt \\ &\leq \|x\|^2 + \|y\|^2 + 2 \left(\int_{-1}^1 |(x(t), x'(t))|^2 dt \right)^{1/2} \cdot \left(\int_{-1}^1 |(y(t), y'(t))|^2 dt \right)^{1/2} \\ &= \|x\|^2 + \|y\|^2 + 2\|x\|\|y\| \\ &= (\|x\| + \|y\|)^2, \end{aligned}$$

which verifies the triangle inequality. Hence $\|\cdot\|$ is a norm.

To show that X is not complete under $\|\cdot\|$, we note that $x_k \rightarrow x$ pointwisely and $x'_k \rightarrow x'$ a.e. (everywhere in $[-1, 1]$ except at $t = 0$). Moreover $|x_k(t) - x(t)|^2, |x'_k(t) - x'(t)|^2 \leq 1$ for all t but $t = 0$. Hence by the dominated convergence theorem, we know

$$\int_{-1}^1 |x_k(t) - x(t)|^2 + |x'_k(t) - x'(t)|^2 dt \rightarrow 0.$$

Therefore $\{x_k\}$ is Cauchy in X under $\|\cdot\|$, but the limit $x \notin C^1([-1, 1])$. Hence X is not complete. □

3. Let $\|\cdot\|_1$ and $\|\cdot\|_2$ be two norms defined on $C([0, 1])$ as follows:

$$\|f\|_1 = \left(\int_0^1 |f(x)|^2 dx \right)^{1/2}, \quad \|f\|_2 = \left(\int_0^1 (1+x)|f(x)|^2 dx \right)^{1/2}.$$

Show that these two norms are equivalent in $C([0, 1])$.

Proof. It is easy to verify that $\|f\|_1 \leq \|f\|_2 \leq \sqrt{2}\|f\|_1$. Hence the two norms are equivalent. $\square$

4. Let $X := \{f \in C([0, \infty)) : \exists M > 0, \text{ s.t. } |f(x)| \leq M, \forall x \geq 0\}$. For any $a > 0$, define

$$\|f\|_a := \left(\int_0^\infty e^{-ax} |f(x)|^2 dx \right)^{1/2}.$$

Show that $\|\cdot\|_a$ is a norm on X .

Proof. Define $A : C([0, \infty)) \rightarrow C([0, \infty))$ where $(Af)(x) = e^{-ax/2}f(x)$ for every $f \in C([0, \infty))$. Then we know $\|f\|_a = \|Af\|_2$. It is easy to verify that $\|\cdot\|_a$ is positive definite and homogeneous. Moreover,

$$\|f + g\|_2 = \|A(f + g)\|_2 = \|Af + Ag\|_2 \leq \|Af\|_2 + \|Ag\|_2 = \|f\|_a + \|g\|_a,$$

which verifies the triangle inequality. Hence $\|\cdot\|_a$ is a norm. $\square$

5. Consider the same setting as in Problem 4. Suppose $b > a > 0$, show that $\|\cdot\|_b$ and $\|\cdot\|_a$ are not equivalent. [Hint: it suffices to show that there exists $\{f_k\}$ such that $\|f_k\|_a \geq k\|f_k\|_b$ for all k sufficiently large. For example, consider $f_k(x)$ set to $e^{bx/2}$ on $[0, k]$, $-e^{bk/2}(x - k - 1)$ on $(k, k + 1]$, and 0 on $(k + 1, \infty)$.]

Proof. We can see that

$$\begin{aligned} \|f_k\|_a^2 &= \int_0^k e^{-ax} e^{bx} dx + \int_k^{k+1} e^{-ax} e^{bk}(x - k - 1)^2 dx \geq \int_0^k e^{(b-a)x} dx \geq \frac{e^{(b-a)k} - 1}{b-a}, \\ \|f_k\|_b^2 &= \int_0^k e^{-bx} e^{bx} dx + \int_k^{k+1} e^{-bx} e^{bk}(x - k - 1)^2 dx \leq k + 1. \end{aligned}$$

Since $b > a$, we know $\frac{e^{(b-a)k} - 1}{(b-a)(k+1)k^2} \rightarrow \infty$ as $k \rightarrow \infty$. Hence there exists $K \in \mathbb{N}$, such that

$$\frac{\|f_k\|_a}{\|f_k\|_b} \geq \left(\frac{e^{(b-a)k} - 1}{(b-a)(k+1)} \right)^{1/2} \geq k, \quad \forall k \geq K.$$

Therefore $\|\cdot\|_b$ and $\|\cdot\|_a$ are not equivalent. $\square$

6. Let $(X, \|\cdot\|)$ be a B^* space. Show that X is complete if and only if $\sum_{k=1}^\infty x_k$ converges whenever the sequence $\{x_k\}$ satisfies $\sum_{k=1}^\infty \|x_k\| < \infty$. [Hint: use the fact that a Cauchy sequence is convergent if and only if a subsequence is convergent.]

Proof. ($\Rightarrow$) Suppose X is complete and $\sum_{k=1}^\infty \|x_k\| < \infty$. Let $s_k = \sum_{i=1}^k x_i$. Then

$$\|s_{n+p} - s_n\| = \left\| \sum_{k=n+1}^{n+p} x_k \right\| \leq \sum_{k=n+1}^{n+p} \|x_k\| \rightarrow 0$$

as $n \rightarrow \infty$ and for any $p \in \mathbb{N}$. Hence $\{s_k\}$ is Cauchy and thus there exists $s \in X$ such that $s_k \rightarrow s$. This implies that $\sum_{k=1}^\infty x_k$ is convergent (with limit s).

($\Leftarrow$) Let $\{s_n\}$ be Cauchy in X . Then for any $k \in \mathbb{N}$, there exists $n_k \in \mathbb{N}$, such that

$$\|s_{n+p} - s_n\| < \frac{1}{2^k}, \quad \forall n \geq n_k, p \in \mathbb{N}.$$

WLOG, we assume $n_1 < n_2 < \dots$, and denote $x_k := s_{n_{k+1}} - s_{n_k}$ and $s_{n_0} = 0$. Thus

$$\sum_{k=1}^{\infty} \|x_k\| \leq \sum_{k=1}^{\infty} \frac{1}{2^k} < \infty.$$

Thus $\{s_{n_k} = \sum_{i=0}^{k-1} x_i\}$ is a convergent subsequence of $\{s_k\}$. Hence $\{s_k\}$ is convergent. $\square$

7. Let X be a B^* space and $f : X \rightarrow \mathbb{R}$ be convex, i.e.,

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y), \quad \forall \lambda \in [0, 1], \quad x, y \in X.$$

Suppose x^* is a local minimizer of f , i.e., there exists $\delta > 0$ such that $f(x^*) = \min_{x \in B(x^*; \delta)} f(x)$. Show that x^* is a global minimizer of f , i.e., $f(x^*) \leq f(x)$ for all $x \in X$.

Proof. Assume not, i.e., there exists $y \in X$ such that $f(y) < f(x^*)$. Let $\lambda = \frac{\delta}{2\|x^* - y\|}$ and $z = (1 - \lambda)x^* + \lambda y$, then $\|z - x^*\| = \|\lambda(x^* - y)\| = \delta/2 < \delta$. Therefore $z \in B(x^*; \delta)$, and thus

$$f(x^*) \leq f(z) = f((1 - \lambda)x^* + \lambda y) \leq (1 - \lambda)f(x^*) + \lambda f(y) < f(x^*),$$

which is a contradiction. $\square$

8. Let X be a B^* space and M a proper finite dimensional subspace of X . Show that there exists $y \in X$ such that $\|y\| = 1$ and $\|y - x\| \geq 1$ for all $x \in M$.

Proof. Since M is a proper finite dimensional subspace of X , we know there exists $z \in X \setminus M$ and $x^* \in M$, such that

$$0 < \|z - x^*\| = d(z, M) = \min_{x \in M} \|z - x\|.$$

Let $y = \frac{z - x^*}{\|z - x^*\|}$, then it is easy to verify that $\|y\| = 1$ and for any $x \in M$ there is

$$\|y - x\| = \left\| \frac{z - x^*}{\|z - x^*\|} - x \right\| = \frac{\|z - (x^* + \|z - x^*\|x)\|}{\|z - x^*\|} \geq 1,$$

since $x^* + \|z - x^*\|x \in M$. $\square$

9. Let $C_0 := \{x = (\xi_1, \xi_2, \dots) : \lim_{k \rightarrow \infty} \xi_k = 0, \xi_k \in \mathbb{R}, \forall k \in \mathbb{N}\}$, i.e., C_0 is the set of real sequences that converge to 0. Define the norm $\|x\| = \sup_{k \in \mathbb{N}} |\xi_k|$. Let $M := \{x = (\xi_1, \xi_2, \dots) \in C_0 : \sum_{k=1}^{\infty} \frac{\xi_k}{2^k} = 0\}$. Show that M is a closed subspace of C_0 .

Proof. Denote $x^{(n)} = (\xi_1^{(n)}, \xi_2^{(n)}, \dots)$ and $x^* = (\xi_1^*, \xi_2^*, \dots)$. Suppose $x^{(n)} \in M$ and $x^{(n)} \rightarrow x^*$ as $n \rightarrow \infty$. Then

$$\left| \sum_{k=1}^{\infty} \frac{\xi_k^*}{2^k} \right| = \left| \sum_{k=1}^{\infty} \frac{\xi_k^{(n)}}{2^k} - \sum_{k=1}^{\infty} \frac{\xi_k^*}{2^k} \right| \leq \sum_{k=1}^{\infty} \frac{|\xi_k^{(n)} - \xi_k^*|}{2^k} \leq \|x^{(n)} - x^*\| \rightarrow 0,$$

as $n \rightarrow \infty$. Hence $\sum_{k=1}^{\infty} \frac{\xi_k^*}{2^k} = 0$ which means $x^* \in M$. Therefore M is closed. $\square$

10. Consider the same setting as Problem 9. Let $x_0 = (2, 0, 0, \dots)$. Show that

$$\inf_{z \in M} \|x_0 - z\| = 1,$$

but $\|x_0 - z\| > 1$ for all $z \in M$. [Hint: consider $z^{(n)} = (1 - \frac{1}{2^n}, -1, \dots, -1, 0, \dots)$ with the first n terms being nonzero. Remark: this example shows that minimum may not be achieved if M is infinite dimensional.]

Proof. It is easy to verify that $z^{(n)} \in M$ for all $n \in \mathbb{N}$. Moreover $\|x_0 - z^{(n)}\| = 1 + \frac{1}{2^n} \rightarrow 1$ as $n \rightarrow \infty$. Suppose there exists $z = (\xi_1, \xi_2, \dots) \in M$ such that $\|x_0 - z\| \leq 1$. As $z \in M$, we know $\xi_1 = \sum_{k=2}^{\infty} \frac{\xi_k}{2^k}$. Since

$$\|x_0 - z\| = \|(2 - \xi_1, \xi_2, \xi_3, \dots)\| \leq 1,$$

we know $|2 - \xi_1| \leq 1$ and $\sup_{k \geq 2} |\xi_k| \leq 1$. Therefore

$$|\xi_1| = \left| \sum_{k=2}^{\infty} \frac{\xi_k}{2^k} \right| \leq \sum_{k=2}^{\infty} \frac{|\xi_k|}{2^k} \leq \sum_{k=2}^{\infty} \frac{1}{2^k} = \frac{1}{2}.$$

However, on the other hand, $|2 - \xi_1| \leq 1$ implies that $1 \leq \xi_1 \leq 3$, which is a contradiction. Hence $\|x_0 - z\| > 1$ for all $z \in M$.

Combing the results above, we know $\inf_{z \in M} \|x_0 - z\| = 1$. □